

3'end RNASeq using template switching protocol (Updated June 23 NCH)

RNA extraction

- Maxwell RNA extraction
 - o Alternative: Add 3 volumes of LS-Trizol to lysate
 - o Extract RNA using the Zymo MicroSpin Columns
 - o Elute with 6 ul of H₂O

RNA fragmentation and template switching RT

- Prepare fragmentation buffer

Fragmentation buffer

80 mM Tris-HCl pH 7.5

40 mM MgCl₂

NOTE: have more recently been using milder fragmentation buffer conditions with input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl₂ (final concentration 1.25 mM Tris-HCl and 0.625mM MgCl₂)

- Use 500 ng of RNA (whenever possible)
- Bring to 3.5 ul using water
- Add 0.5 ul of fragmentation buffer (10 mM Tris-HCl pH 7.5, 5 mM MgCl₂ final concentration)
- Fragment RNA by incubating 12 min at 95C
- Add
 - 0.5 ul 10 mM dNTPs
 - 0.5 ul 5 uM primer RT1
- Mix and Incubate at 65C for 3 min then cool down to 42 C & hold
- Make a mastermix and add 5.5 ul of the following
 - o 2 uL of SSIV buffer
 - o 2 uL 5M betaine
 - o 0.5 uL 0.1 M DTT
 - o 0.25 uL 40 uM TSO
 - o 0.25 uL 200 mM MgCl₂
 - o 0.25 uL SuperRNasin
 - o 0.25 uL SSIV
- Place the mastermix into a PCR strip and into the cycler, so that it pre-warms 42 C before you start adding it to the samples
- Add mastermix to samples while keeping them at 42 C in the cycler (i.e., don't take the samples out of the cycles while adding the mix)

- Incubate at 42 C for 10min
- Incubate at 50 C for 30min
- Purify using Magbind beads (1.8 ul per 1ul of PCR reaction). Elute in 8uL H2O

Optional: test PCR

- Use 1uL RT product

Indexing PCR

- Prepare the following 20ul indexing PCR reaction for each sample:
 - o 10 uL of 2x Q5
 - o 2ul RT product
 - o 1ul dual indexed primers
 - Use a different index for each sample. Make sure each sample will have a different index if submitting multiple libraries at the same time.
 - o 7 uL of ddH2O
- Incubate for the PCR reaction using the Q5 program
 - o Extension time - 45 seconds, Tm – 65 C
 - o Typically try 10 and 14 cycles
- Move samples to post-PCR room
- Add 2 ul of purple 6X loading buffer to 5 ul of PCR reaction. Keep the rest of the 20 ul
- Load on a 6% TBE gel together of 2 uL of ready to use low molecular weight ladder (NEB)
- Run at 180V for 25 min
- Stain with Sybr Green (1:10000 dilution) in TBE 1X
- Visualise

PCR beads purification

- Mix together approx. 10 ul of each PCR reaction
- Add a suitable amount of Magbind beads (e.g. 0.9 ul per 1ul of PCR reaction) and mix
- Incubate at room temperature for 5 minutes
- Remove supernatant and wash with 500 ul of 70% Ethanol for 30 Sec. Repeat this step
- Air dry the beads
- Resuspend in 30 ul of H2O